



**Politecnico
di Torino**

Politecnico di Torino

Lab report for the course Qubit Electronics

Master degree in Quantum Engineering

Group 1

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Contents

1	Python Script	1
2	Analysis of the Results	3

CHAPTER 1

Python Script

Question: Report the line of code added.

```
1 for idx, V_barrier_2 in enumerate(V_Barrier_Vec_2):
2     d.new_gate_bnd('surface_Plunger_R_X_gate', V_Plunger_Right_Vec[idx], mWF)
3     d.new_gate_bnd('surface_Barrier_M_X_gate', V_barrier_2, mWF)
4
5
6     #-----#
7     #
8     #     Poisson Solver
9     #
10    #-----#
11
12
13    # Create an adaptive-mesh non-linear Poisson solver
14    p = PoissonSolver(d, solver_params=params_poisson, geo_file=path_geo)
15
16    # Self-consistent solution
17    p.solve()
18
19
20    #-----#
21    #
22    #     Schrodinger Solver
23    #
24    #-----#
25    num_states = 5
26
27    # Get the potential energy from the band edge for usage in the Schrodinger
28    # solver
29    d.set_V_from_phi()
30
31    # Create a submesh including only the dot region
32    submesh = SubMesh(d.mesh, dot_region)
33
34    # Create a subdevice for the dot region
35    subdvc = SubDevice(d, submesh)
36
37    # Create a Schrodinger solver
38    schrod_solver = SchrodingerSolver(subdvc, solver_params=params_schrod)
39
40    # Solve Schrodinger's equation
41    schrod_solver.solve()
42
```

```

43 # Output and save single-particle energy levels
44 subdvc.print_energies()
45 energies = subdvc.energies
46 np.save(path_out/f"dqdfdsol_energies_{idx}.np", energies)
47
48 # To save throughout the device: n, p, phi, EC, EV
49 arrays_dict = {"n": d.n/1e6, "p": d.p/1e6, "phi": d.phi,
50 "EC": d.cond_band_edge()/ct.e, "EV": d.vlnce_band_edge()/ct.e}
51 io.save(str(path_hdf5), arrays_dict)
52 #io.save(str(path_vtu), arrays_dict, d.mesh)
53 arrays_dict = {"n": d.n/1e6}
54 io.save(str(path_out / f"n_distribution_{idx}.vtu"), arrays_dict, d.mesh)
55
56 arrays_dict = {"p": d.p/1e6}
57 io.save(str(path_out / f"p_distribution_{idx}.vtu"), arrays_dict, d.mesh)
58
59 arrays_dict = {"phi": d.phi}
60 io.save(str(path_out / f"phi_distribution_{idx}.vtu"), arrays_dict, d.mesh)
61
62 arrays_dict = {"EC": d.cond_band_edge()/ct.e}
63 io.save(str(path_out / f"EC_distribution_{idx}.vtu"), arrays_dict, d.mesh)
64
65 arrays_dict = {"EV": d.vlnce_band_edge()/ct.e}
66 io.save(str(path_out / f"EV_distribution_{idx}.vtu"), arrays_dict, d.mesh)
67
68 # Schrodinger
69 out_dict = {"Ground": np.abs(subdvc.eigenfunctions[:,0])**2,
70 "1st excited": np.abs(subdvc.eigenfunctions[:,1])**2,
71 "2nd excited": np.abs(subdvc.eigenfunctions[:,2])**2,
72 "3rd excited": np.abs(subdvc.eigenfunctions[:,3])**2,
73 "4th excited": np.abs(subdvc.eigenfunctions[:,4])**2,
74 "conf-potential": subdvc.V/ct.e}
75 io.save(str(path_out / f"schrodinger_results_{idx}.vtu"), out_dict, subdvc.mesh)
76 with open(path_out / f"energies_{idx}.txt", 'w') as f:
77     f.write('Energy eigenvalues [eV]\n')
78     f.write(str(subdvc.energies / ct.e))
79
80
81 energy_arr[idx,1:] = subdvc.energies/ct.e
82 subdvc.print_energies()
83 delta_E = (subdvc.energies[1] - subdvc.energies[0]) / ct.e # in eV
84 splitting_arr[idx, 0] = V_barrier_2
85 splitting_arr[idx, 1] = delta_E
86
87 np.savetxt(path_splitting, splitting_arr, header="V_barrier_2 [V]    Delta_E (E1 -
88 E0) [eV]")
89 np.save(path_out / "splitting_array.npy", splitting_arr)
90 np.save(path_out / "energy_levels.npy", energy_arr)
91 np.savetxt(path_out / "energy_levels.txt", energy_arr, header="V_barrier_2 [V], E0
, E1, E2, E3, E4 [eV]")

```

We added the Schrodinger and Poisson solvers inside the for loop. Then, for every value of the tunnel barrier, we calculate the energy difference between the two energy levels (ground state and first excited). The results are then saved in a .txt file.

Question: Which parameters related to the structure differ from the previous laboratory?

The parameter that changed at every simulation was the value of the tunnel barrier, ranging from 0.37 to 0.42 V. The structure of the device is the same, as regards both the dimensions and the materials and doping of the various regions.

CHAPTER 2

Analysis of the Results

We report the values of the energy eigenvalues obtained for each tunnel barrier voltage.

V_{barrier} [V]	E_0 [eV]	E_1 [eV]	E_2 [eV]	E_3 [eV]	E_4 [eV]
0.370	0.01229490	0.01229732	0.01497502	0.01497970	0.01557596
0.380	0.01196489	0.01196713	0.01467971	0.01468529	0.01522736
0.390	0.01162993	0.01163234	0.01437929	0.01438609	0.01486293
0.400	0.01128982	0.01129278	0.01407358	0.01408196	0.01448330
0.410	0.01094440	0.01094821	0.01376239	0.01377276	0.01408938
0.420	0.01059344	0.01059840	0.01344548	0.01345831	0.01368185

Table 2.1: Energy eigenvalues (E_0 to E_4) as a function of tunnel barrier voltage V_{barrier} .

By increasing the tunnel barrier voltage, we observe an increase in the energy level splitting between the first two levels.¹ This behavior is well fitted by an exponential function.

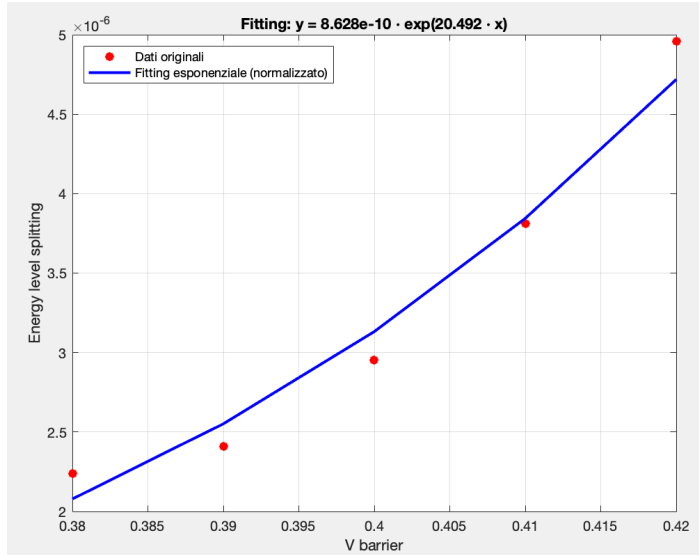


Figure 2.1

¹The fit presented excludes the eigenvalues obtained with the lowest tunnel barrier value of 0.37 V. It has been reported that, at this barrier height, the double quantum dot enters a special regime in which QTCAD produces unreliable results.